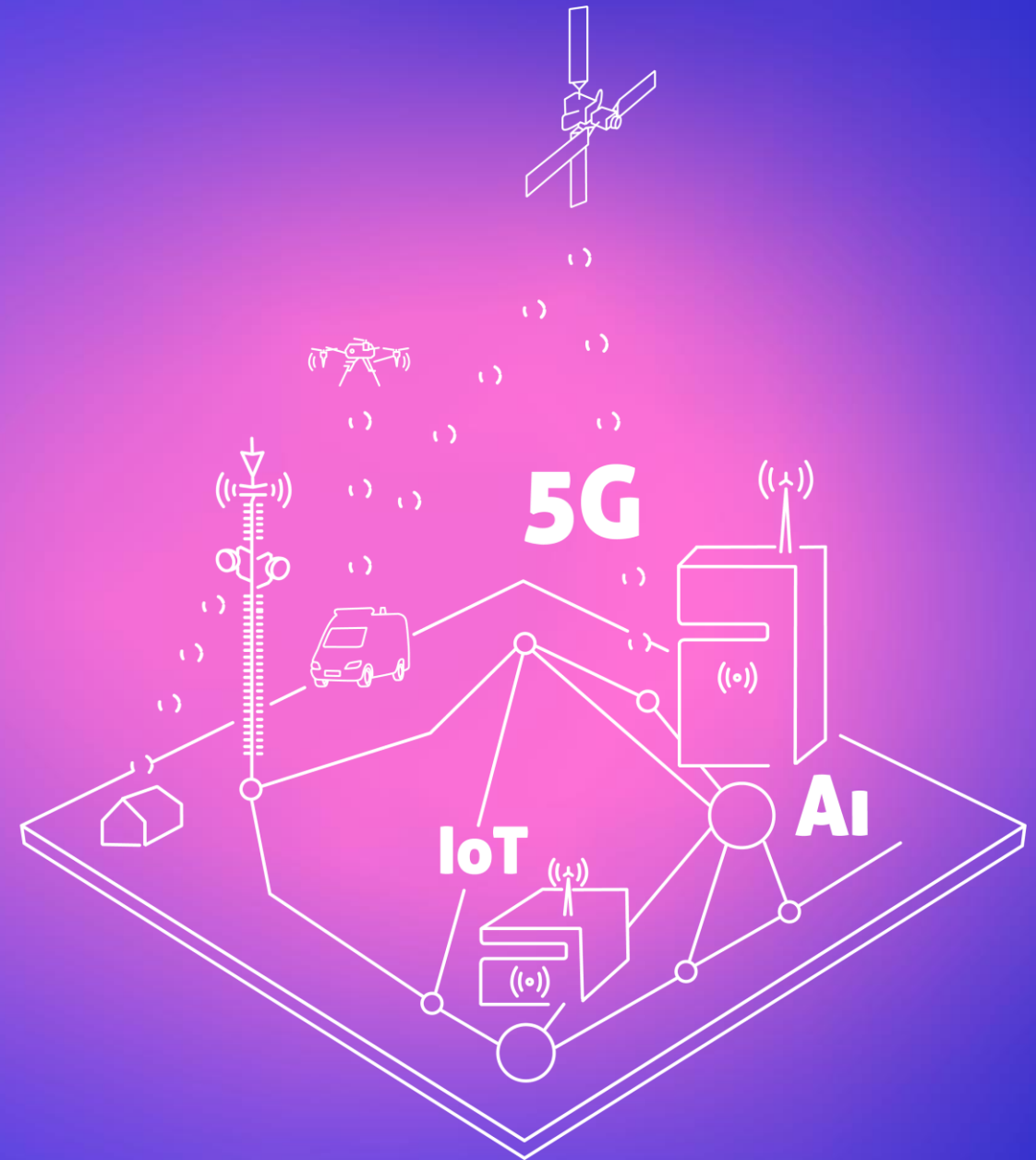




Introduction: CE Structure

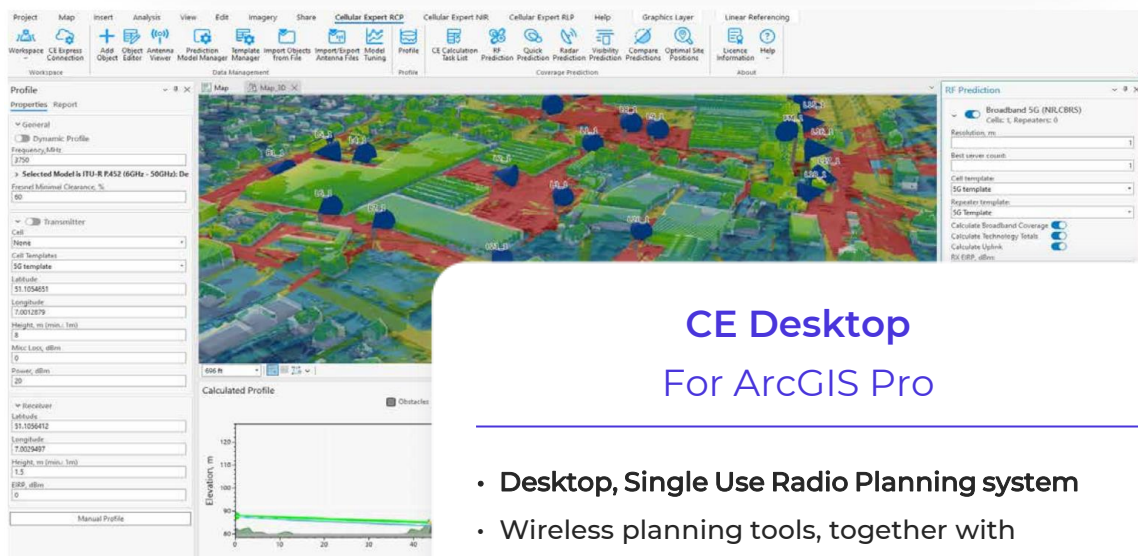
www.cellular-expert.com



CE Products and Solutions

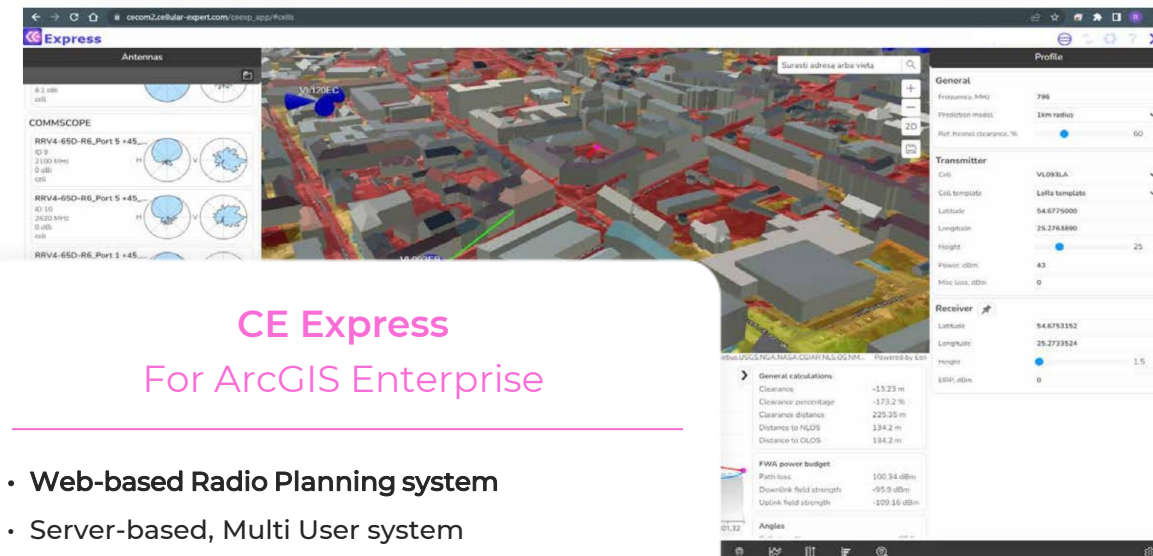


ArcGIS System Ready
Specialty



CE Desktop For ArcGIS Pro

- Desktop, Single Use Radio Planning system
- Wireless planning tools, together with ArcGIS Pro functionality
- Can be used as a client of CE Express system



CE Express For ArcGIS Enterprise

- Web-based Radio Planning system
- Server-based, Multi User system
- Includes CE Inventory3D
- Dashboard of Network Coverage Statistics

CE Customized solutions

ArcGIS

CE COTS

CE Pro



Project

Map

Insert

Analysis

View

Edit

Imagery

Share

Cellular Expert RCP

Cellular Expert NIR

Cellular Expert RLP

Help

Graphics

Graphics Layer

Workspace

CE Express Connection

Add Object

Object Editor

Antenna Viewer

Prediction Model Manager

Template Manager

Import Objects from File

Import/Export Antenna Files

Model Tuning

Profile

CE Calculation Task List

RF Prediction

Quick Prediction

Radar Prediction

Visibility Prediction

Compare Predictions

Optimal Site Positions

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Profile

Coverage Prediction

Contents

Search

Drawing Order

☐ Minimum Receiver H...

Value

< 5m

5 - 20m

20 - 50m

> 50m

☐ Line of Sight

Value

0

1

☒ Clearance

Value

< -10m

-10 - 0m

0 - 5m

> 5m

☐ Best Server

Map

Map_3D

Scene

Map1

Map2

1:2,843

583,658.83E 6,061,098.17N m

Selected Features: 0

Calculated Profile

Distance: 32.22 m

Fresnel (Primary): 133.81 m

Fresnel (Secondary - 60% Clearance): 134.91 m

Profile: 136.56 m

Obstacles: 124.93 m

Elevation: 111.35 m

Clutter: 111.35 m

Show Table

Change Col

Profile

Properties

Report

☐ Transmitter

Cell

None

Cell Templates

5G template

Latitude

54.679167

Longitude

25.290278

Height, m (min.: 1m)

38

Misc Loss, dBm

0

Power, dBm

40

Receiver

Latitude

54.6790514

Longitude

25.2917103

Cellular Expert for ArcGIS Pro Architecture



Cellular Expert for ArcGIS Pro RCP | RLP | EMF

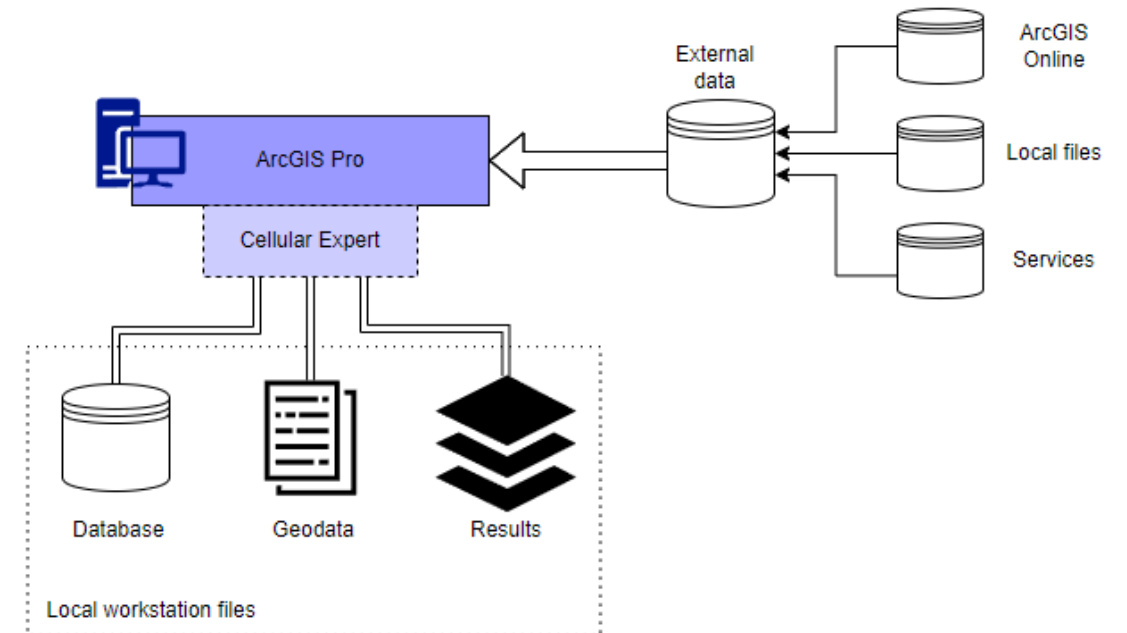
ArcGIS Pro 3.1 and
above

Basic or higher license

Additional extensions are not
required

Local Database

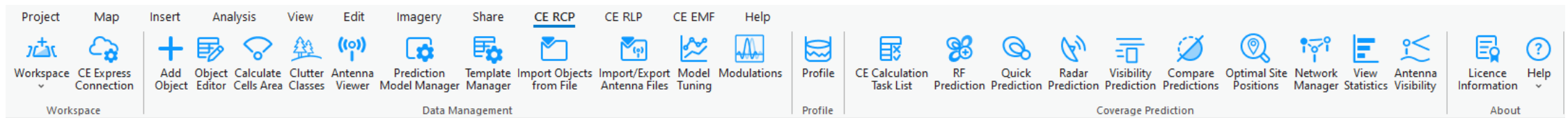
All files are saved in local disk



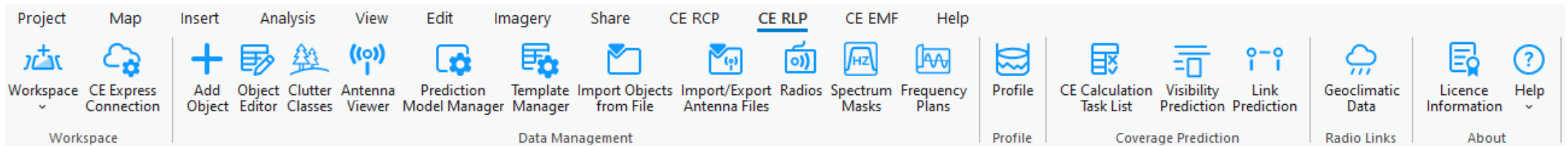
Cellular Expert for ArcGIS Pro License Structure



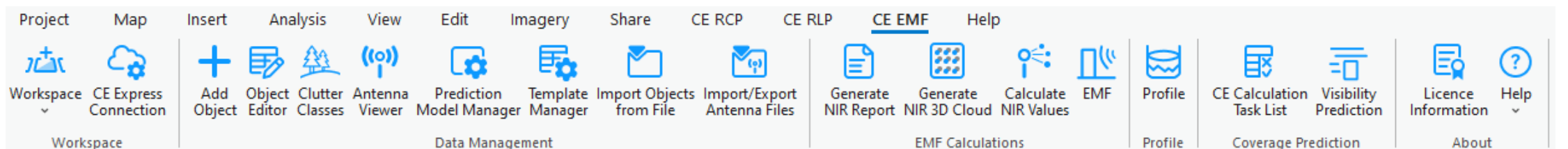
- RCP – Radio Coverage Prediction



- RLP – Radio Link Prediction

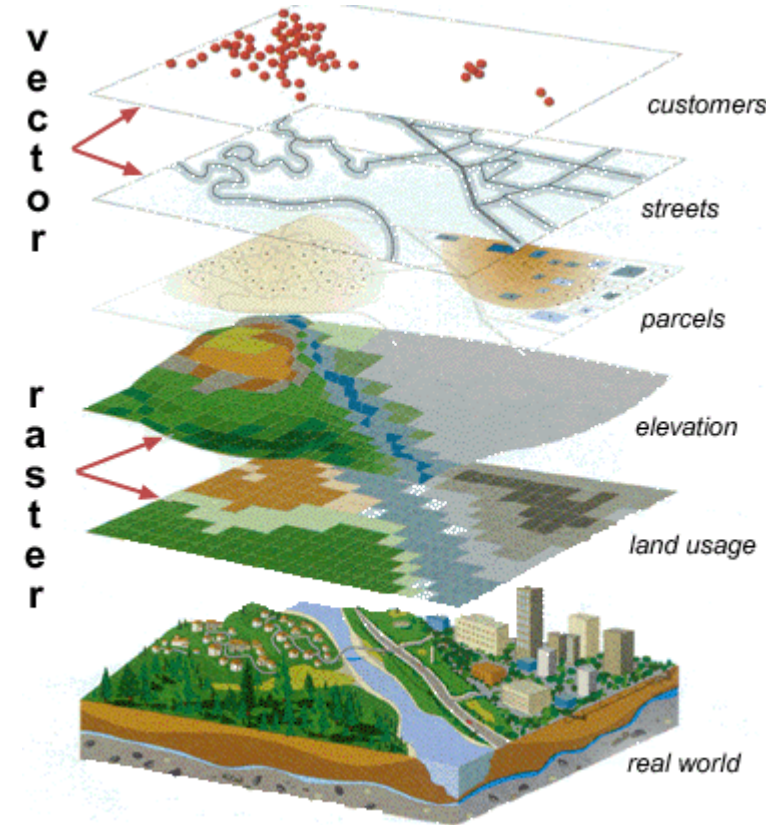


- EMF – Electromagnetic Field



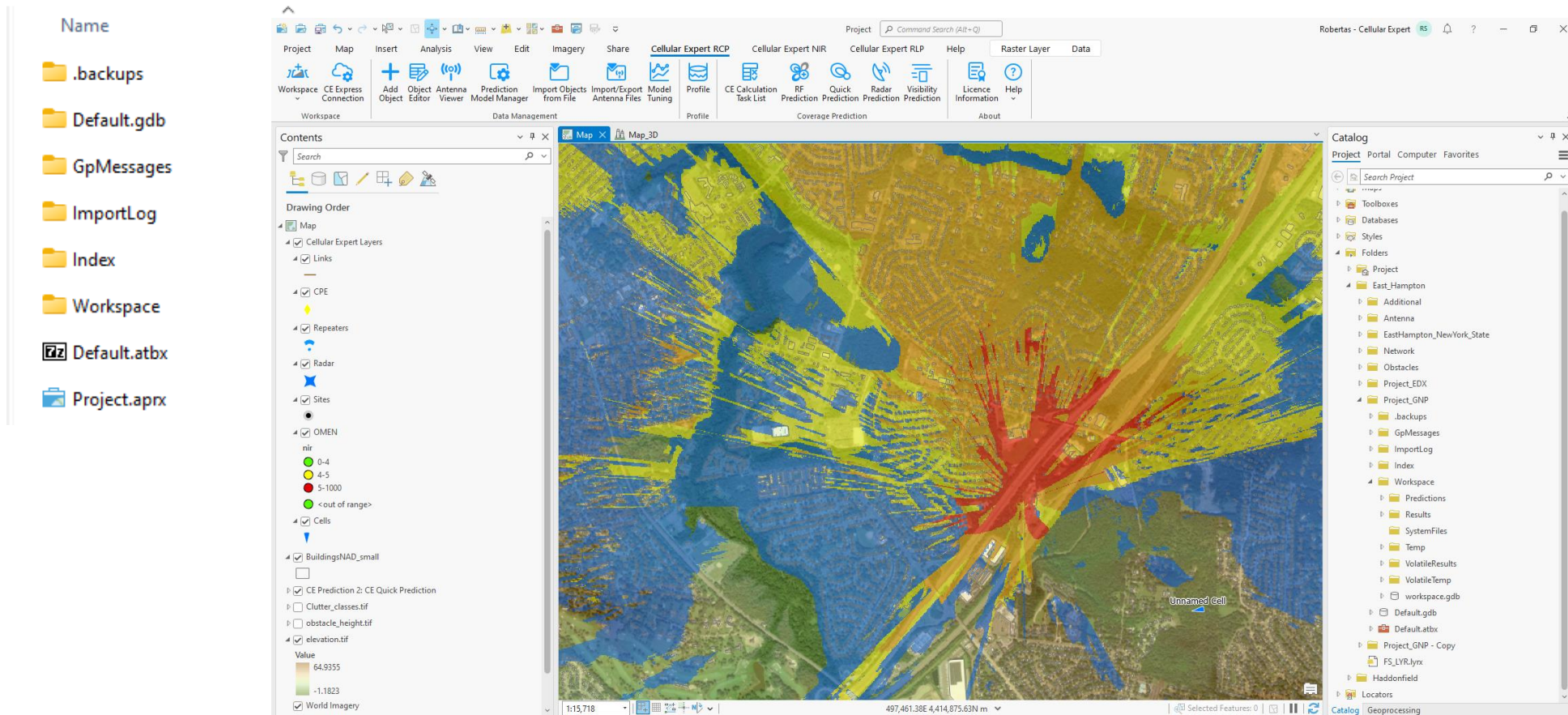
Geospatial Information

- ✓ Field measurements
- ✓ Network coverage
- ✓ Network data
- ✓ Demography
- ✓ Land cover / use
- ✓ Obstacles
- ✓ Elevation
- ✓ Surface



Project files

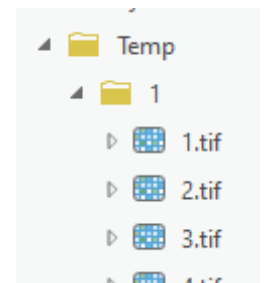
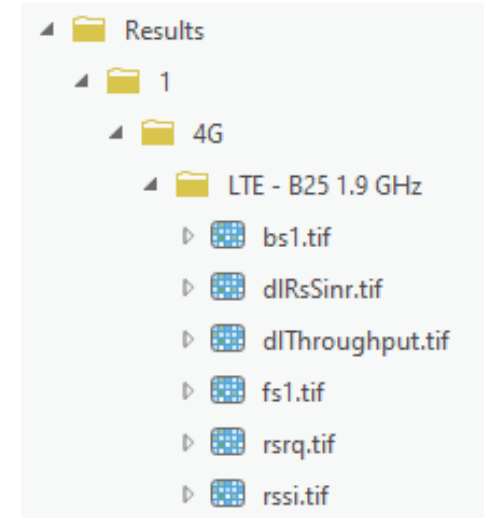
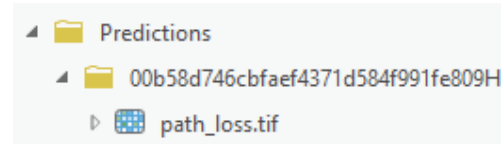
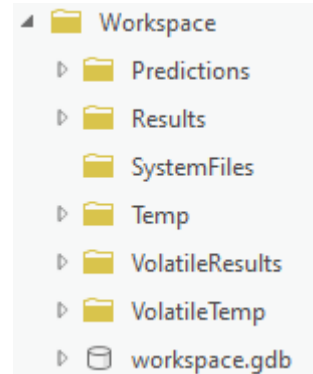
Project > Save Project/Save Project As



Cellular Expert Project Structure



- Predictions
- Results
- SystemFiles
- Temp
- VolatileResults
- VolatileTemp
- Workspace.gdb



Workspace database files



workspace.gdb

Cellular_Expert_Dataset

Cells

CPE

Links

OMEN

OMEN1

Radar

Repeaters

Sites

Antennas

CE_LAYERS

CE_TABLEDATA

CE_WORKSPACE

CellTemplates

ClutterLoss

CPETemplates

Cqi4G

Cqi5G

LinkTemplates

MimoThroughput

model_oneslope_plus_ske

model_unimacro

prediction_models

RadarTemplates

RepeaterTemplates

Contents

Search

Drawing Order

Map

Cellular Expert Layers

Links

CPE

Repeaters

Radar

Sites

OMEN

nir

0-4

4-5

5-1000

<out of range>

Cells

CE Prediction 33: CE Calculation

5G

5G 3500

RSSI

Field Strength 3

Field Strength 2

Field Strength 1

Value

-155 - -119

Project30

Command Search (Alt+Q)

Robertas - Cellular Expert RS

Project

Map

Insert

Analysis

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Cellular Expert RLP

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Map_3D

Layout

MS182794

MS182841

MS032813

MS027614

MS099214

MS159395

MS159396

MS185362

MS174407

MS048828

MS099211

1:22,502

580,760.44E 6,061,907.75N m

Selected Features: 3

Antenna Viewer

Attribute Filter

Antenna Type:

Sector

Selected Antenna: 10P-2L8M-B5-V2_Port 3 +45_01DT_1800

Object ID

Manufactur

Model

Frequency

Gain, dBi

1

COMMSCOP

10P-2L8M-B

1800

16.41

2

WiFi

Antenna 2.4

900

8.1

3

WiFi

Antenna 5G

900

8.1

4

COMMSCOP

DB583-Y_0

903

5.10999999

5

COMMSCOP

HWX-6516D

2300

17.97

6

COMMSCOP

RRV4-65D-F

700

0

7

COMMSCOP

RRV4-65D-F

1800

0

8

COMMSCOP

RRV4-65D-F

2100

0

9

COMMSCOP

RRV4-65D-F

2620

0

10

COMMSCOP

SBNHH-1D6

704

14.41

11

0

12

0

Antenna Horizontal P

Antenna Vertical Patte

Horizontal Attenuation, dBm

Vertical Attenuation, dBm

Angle, °

Horizontal Att

Vertical Attenu

0

0.1

0.23

1

0.07

0

2

0.05

0.04

3

0.04

0.36

4

0.02

0.98

5

0.01

1.96

6

0.01

3.35

7

0

5.2

8

0

7.56

9

0

10.48

10

0.01

14.12

11

0.03

18.65

12

0.05

22.05

13

0.07

20.66

Object Editor

Move Objects

Duplicate Objects

Unassigned Cells

MS182794 (Object ID: 316)

MS182795 (Object ID: 317)

MS182796 (Object ID: 318)

View From Cell

0

meters above its location

Technology:

5G

Duplex Mode:

FDD

Frequency, MHz:

3500

Frequency group:

5G 3500

Tilt, deg:

0

Selected Antenna: 10P-2L8M-B5-V2_Port 3 +45_01DT_1800

Object ID

Manufactur

Model

Frequency

Gain, dBi

1

COMMSCOP

10P-2L8M-B

1800

16.41

2

WiFi

Antenna 2

900

8.1

3

WiFi

Antenna 5

900

8.1

4

COMMSCOP

DB583-Y_0

903

5.10999999

5

COMMSCOP

HWX-6516

2300

17.97

6

COMMSCOP

RRV4-65D

700

0

7

COMMSCOP

RRV4-65D

1800

0

Save Changes

Dismiss

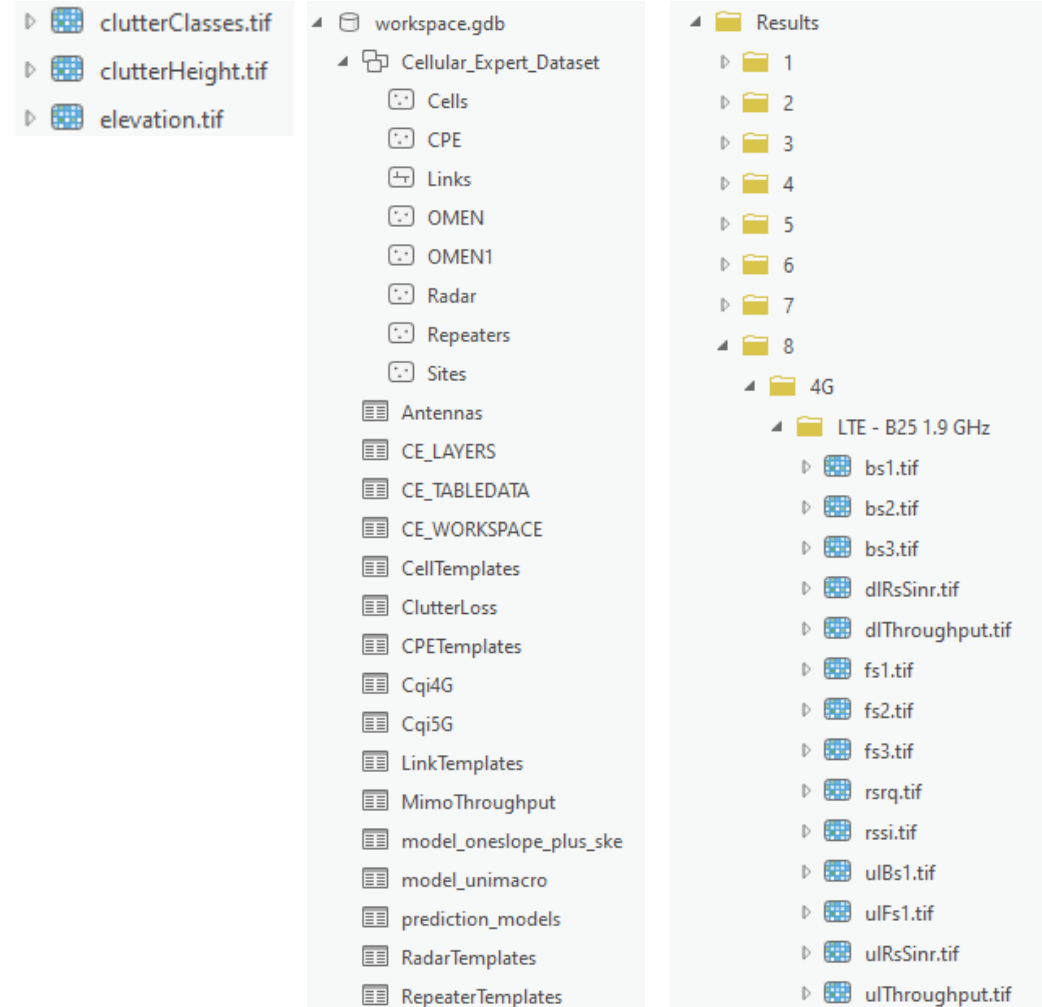
Catalog

Geoprocessing

Object Editor

Environment

- Geographic data
- Cellular Expert Workspace
- Results



Cellular Expert Workspace



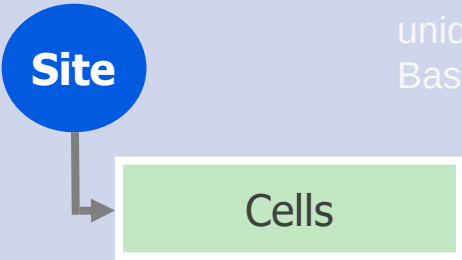
The screenshot shows the 'Workspace Properties' dialog box with the 'Parameters' tab selected. The 'Workspace' menu is open, showing options: 'Create', 'Open', 'Remove', 'Upgrade', and 'Properties'. The 'Parameters' table is divided into three sections: 'CE Server', 'Project Paths', and 'Project Settings'.

Name	Value
CE Server	
ArcGIS Portal URL	
CE Server API	
CE Server Environment	
CE Server URL	
CE Server Workspace	
CE Server Workspace Id	
Project Paths	
Batch Logs Path	
Calculation Path	C:\CE\VinidiusDemo\Project30\Workspace1\Temp
Calculation Tasks Data Path	C:\CE\VinidiusDemo\Project30\Workspace1\SystemFiles\calculation_tasks.db
Geoclimatic Data Path	
Geodata Folder Path	C:\CE\VinidiusDemo\Vinidius
Prediction Path	C:\CE\VinidiusDemo\Project30\Workspace1\Predictions
Result Path	C:\CE\VinidiusDemo\Project30\Workspace1\Results
Volatile Calculation Path	C:\CE\VinidiusDemo\Project30\Workspace1\VolatileTemp
Volatile Result Path	C:\CE\VinidiusDemo\Project30\Workspace1\VolatileResults
Volatile Tasks Data Path	C:\CE\VinidiusDemo\Project30\Workspace1\SystemFiles\volatile_tasks.db
Project Settings	
Link templates: Use existing sectors	No
Load impedance (Ohms)	50

At the bottom of the dialog are buttons for 'OK', 'Cancel', and 'Help'.

- Network Data
- Equipment Data
- Modelling Settings

Network Data Structure

 Cell – logical information about sector: a set of channels	RF prediction does not require Site object. Cells can be created without Site object. Cell still has SiteID value, which can be define in the attributes. SiteID is used for Carrier Aggregation.
 Site – location point with unique identifier: Base station	Site and Cell is connected through SiteID value. SiteID must be Integer. Site name is defined for Site object.

Object Editor

Move Objects

Duplicate Objects

Unassigned Cells

MS182794 (Object ID: 316)

MS182795 (Object ID: 317)

MS182796 (Object ID: 318)

View From Cell

0 meters above its location

Cell Properties

Object ID:

316

Name:

MS182794

Latitude:

54.693900000164135

Longitude:

25.242270999856128

X:

580084.0151000004

Y:

6062650.9604

Z:

133.0000015258789

Height above ground, m:

31.1

Ground altitude, m:

101.9000015258789

Azimuth, deg:

50

Template:

Optional Properties

Site ID

1

Network name:

Technology:

5G

Duplex Mode:

FDD

Frequency, MHz:

3500

Frequency group:

5G 3500

Tilt, deg:

0

Selected Antenna: 10P-2L8M-B5-V2_Port 3 +45_01DT_1800

Object ID	Manufactu	Model	Frequency	Gain, dBi
1	COMMISC	10P-2L8M	1800	16.41
2	WIFI	Antenna	900	8.1
3	WIFI	Antenna	900	8.1

4	COMMISC	DB583-Y_	903	5.109999
5	COMMISC	HWX-651	2300	17.97
6	COMMISC	RRV4-65	700	0
7	COMMISC	RRV4-65	1800	0
8	COMMISC	RRV4-65	2100	0
9	COMMISC	RRV4-65	2620	0
10	COMMISC	SBNHH-1	704	14.41
11		Typical 3	0	0
12		Typical 6	0	0
13		Typical 9	0	0
14		Typical 9	0	0
15		Typical 0	0	0

View Selected Antenna Pattern

Power, dBm:

65

Antenna gain, dBi:

0

Misc loss, dB:

0

Bandwidth, MHz:

40

Noise figure, dB:

6

Downlink duplex factor:

1

Subcarrier spacing, kHz:

30

Tx mimo:

1

Rx mimo:

1

Active antenna effect:

0

Cell load:

30

Carriers:

Selected model is ITU R. P452: Default

ITU R. P452

Default

UniMacro

Default

- 800 MHz
- 1800 MHz
- 2100 MHz

14

Site

Object Editor

Move Objects Duplicate Objects

Sites

- Site A (Object ID: 1)

Site Properties

Object ID: 1

Name: Site A

Latitude: 54.701324726471995

Longitude: 25.27575853402319

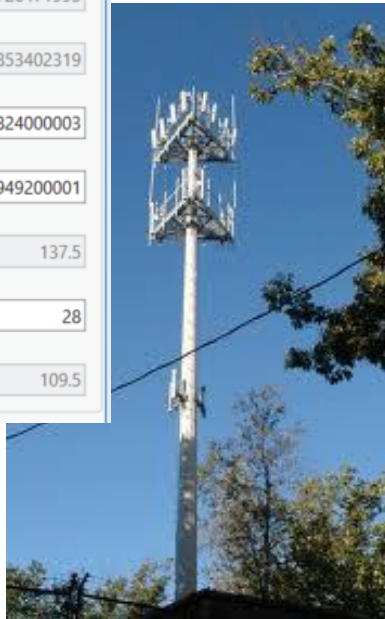
X: 582227.6824000003

Y: 6063515.949200001

Z: 137.5

Height above ground, m: 28

Ground altitude, m: 109.5

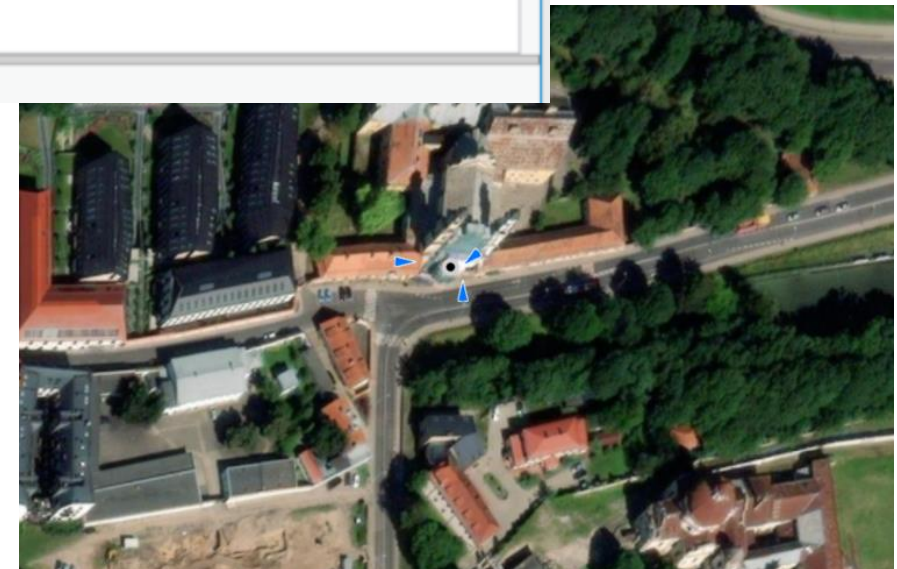


Object Editor

Move Objects Duplicate Objects

Sites

- ST 00H5 (Object ID: 2)
 - Cells
 - MS099211 (Object ID: 266)
 - MS099212 (Object ID: 267)
 - MS099213 (Object ID: 268)



Antenna

Antenna Viewer

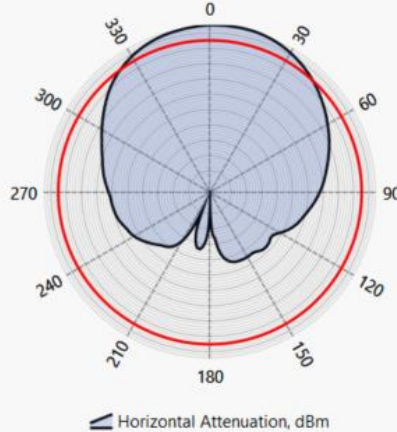
> Attribute Filter ⓘ

Antenna Type: Sector

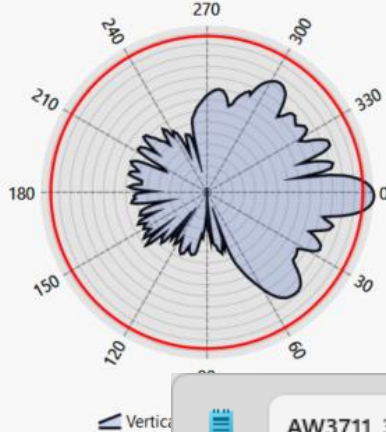
Selected Antenna: 10P-2L8M-B5-V2_Port 3 +45_01DT_1800

Object ID	Manufacturer	Model	Frequency, MHz	Gain, dBi
1	COMMSCOPE	10P-2L8M-B5-V2_P	1800	16.41
2	WiFi	Antenna 2.4GHz 360	900	8.1
3	WiFi	Antenna 5GHz 360	900	8.1
4	COMMSCOPE	DB583-Y_00DT_09K	903	5.1099999999999998
5	COMMSCOPE	HWX-6516DS1-VTM	2300	17.97
6	COMMSCOPE	RRV4-65D-R6_Port	700	0
7	COMMSCOPE	RRV4-65D-R6_Port	1800	0
8	COMMSCOPE	RRV4-65D-R6_Port	2100	0
9	COMMSCOPE	RRV4-65D-R6_Port	2620	0
10	COMMSCOPE	SBNHH-1D65B_Port	704	14.41
11		Typical 33deg H and 0	0	0
12		Typical 65deg H and 0	0	0

Antenna Horizontal Pattern



Antenna Vertical Pattern



Angle, °	Horizontal Attenuation, dBm	Vertical Attenuation, dBm
0	0.1	0.23
1	0.07	0
2	0.05	0.04
3	0.04	0.36
4	0.02	0.98
5	0.01	1.96
6	0.01	3.35
7	0	5.2
8	0	7.56
9	0	10.48
10	0.01	14.12
11	0.03	18.65

AW3711_3!

File Edit View

```
NAME Antenna Patterns
MAKE Kathrein
FREQUENCY 3500
GAIN 20.2dBi
HORIZONTAL 360
0.0 0.0
1.0 0.0
2.0 0.1
3.0 0.2
4.0 0.2
5.0 0.3
6.0 0.5
```

Ln 32, Col 9 100% Windows (CRLF) UTF-8

CE Path Loss models (10kHz - 350 GHz)

1. CEC ITU-R Model (100MHz – 6GHz) is a combination model intended for use in a variety of different radiocommunication systems which is derived explicitly from ITU-R path loss modelling methods as follows:
 - a. Receive antenna in LOS condition – path loss calculated as FSL based on Recommendation ITU-R P.525 ([ref URL](#));
 - b. Receive antenna in OLOS condition – total path loss modelled as a combination of basic FSL calculated based on Recommendation ITU-R P.525 ([ref URL](#)) and clutter loss calculated based on Recommendation ITU-R P.2108 ([ref URL](#));
 - c. Receive antenna in NLOS condition – path loss as a combination of basic FSL calculated based on Recommendation ITU-R P.525 ([ref URL](#)), additional losses due to diffraction calculated based on Recommendation ITU-R P.526 ([ref URL](#)) and the clutter losses calculated based on Rec. ITU-R P.2108 ([ref URL](#)).
2. ITU-R P.452 Model (6GHz – 50GHz) is provided as a universally applicable model with very wide frequency range from 0.1-50 GHz. Its implementation is based on the methodology described in the Recommendation ITU-R P.452 ([ref URL](#)). This model does not provide for definition of OLOS visibility condition; instead it considers clutter as part of general obstacles category and accordingly distinguishes only two radio visibility cases:
 - a. Receive antenna in LOS condition – path loss modelled based on FSL principle;
 - b. Receive antenna in NLOS condition – total path loss modelled using a combination of basic transmission losses and losses due to diffraction.
3. LOS ITU-R P.525 Model (6GHz – 100GHz) is the FSL path loss calculated based on method in Recommendation ITU-R P.525 ([ref URL](#)). As such it could be used for modelling of radio links where LOS is considered a necessary condition, e.g., for Fixed (Point-to-Point) Links or Mobile Systems in mmWave bands.
4. UniMacro Model (400MHz – 3GHz) is the CE's proprietary combination model developed over the years of practical experience with the operational planning of cellular mobile networks in the frequency ranges from 400-2600 MHz. It had been fine tuned to produce coverage predictions that are most closely aligned with what could be expected to be experienced by the actual mobile network users in the field. The model will model different path losses depending on radio visibility conditions as follows:
 - a. Receive antenna in LOS condition – path loss modelled based on FSL principle;
 - b. Receive antenna in OLOS condition – path loss modelled using Extended Hata (Open Area) model with additional clutter loss calculated based on Recommendation ITU-R P.2108 ([ref URL](#));
 - c. Receive antenna in NLOS condition – path loss modelled using Extended Hata model with additional losses due to diffraction calculated based on Recommendation ITU-R P.526 ([ref URL](#)) as well as clutter losses based on Rec. ITU-R P.2108 ([ref URL](#)).
5. ITU-R P.368 Model (10kHz – 30MHz)

CE prediction models

Prediction Model Manager

- ▶ ITU-R P.452 (6GHz - 50GHz)
- ▶ UniMacro (400MHz - 3GHz)
- ▶ CEC ITU-R (100MHz - 6GHz)
- ▶ LOS ITU-R P.525 (6GHz - 100GHz)
- ▶ ITU-R P.368 (10kHz - 30MHz)

Prediction Model Manager

▶ ITU-R P.452 (6GHz - 50GHz)
▶ UniMacro (400MHz - 3GHz)
▲ CEC ITU-R (100MHz - 6GHz)
 Default
 1km radius
 2km radius
 3km radius

Object ID:
1

Configuration name:
Default

Radius, km:
5

Effective Earth radius, km:
8500

Receiver height, m:
1.5

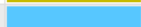
Offset coefficient, dB:
32

Distance coefficient:
20

Distance coefficient obstructed:
40

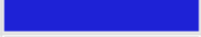
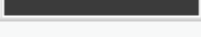
Frequency coefficient:
20

▼ Clutter class:

Clutter name	Color
Buildings	
Forest	
Dense forest	
Urban	
Dense urban	
Bare ground	
Crops	
Water	
Road	

Apply OK Cancel

▼ Clutter class: Buildings

Clutter name	Color
Buildings	
Forest	
Dense forest	
Urban	
Dense urban	
Bare ground	
Crops	
Water	
Road	

Nominal distance, m:
1

Diffraction loss coefficient:
1

Penetration loss offset, dB:
20

Penetration loss distance coefficient:
1.1

Penetration loss frequency coefficient:
0.58

Receiver point loss offset, dB:
0

Apply OK Cancel

Other

clutter						
Field: Add Calculate Selection: Select By Attributes Zoom To Switch						
OBJECTID *	clutter_class_name	clutter_class_id	height	nominal_distance	surfa	
1	1	water	1	0	27	
2	2	trees	2	20	27	
3	3	flooded vegetation	4	3	27	
4	4	crops	5	2		
5	5	built area	7	2		
6	6	bare ground	8	0		
7	7	snow	9	0		
8	8	rangeland	11	3		
Click to add new row.						

CellTemplates						
Field: Add Calculate Selection: Select By Attributes Zoom To Switch						
OBJECTID *	Cell Template Name	Bandwidth	Noise Figure	Subcarrier Spacing	TX MIMO	
1	1	Default (4G 800 Band...	20	6	15	1
2	2	GSM 900 MHz	0.015	6	15	1
3	3	TETRA 400MHz Omni	0.015	6	15	1
4	4	TETRA 400MHz Sector	0.015	6	15	1
5	5	AMI 900MHz Omni	0.015	6	15	1
6	6	LoRa 900MHz Omni	0.015	6	15	1
7	7	LoRa 900MHz Sector	0.015	6	15	1
8	8	3G 900MHz	3.85	6	15	1
9	9	3G 1800MHz	5	6	15	1

RainRegionsITU																	
Field: Add Calculate Selection: Select By Attributes Zoom To Switch																	
OBJECTID *	Percentage	A	B	C	D	E	F	G	H	J	K	L	M	N	P	Q	
1	50	0.001	22	32	42	42	70	78	65	83	55	100	150	120	180	250	170
2	51	0.003	14	21	26	29	41	54	45	55	45	70	105	95	140	200	142
3	52	0.01	8	12	15	19	22	28	30	32	35	42	60	63	95	145	115
4	53	0.03	5	6	9	13	12	15	20	18	28	23	33	40	65	105	96
5	54	0.1	2	3	5	8	6	8	12	10	20	12	15	22	35	65	72
6	55	0.3	0.8	2	2.8	4.5	2.4	4.5	7	4	13	42	7	11	15	34	49
7	56	1	0.1	0.5	0.7	2.1	0.6	1.7	3	2	8	1.5	2	4	5	12	24
Click to add new row.																	

Cqi4G						
Field: Add Calculate Selection: Select By Attributes						
OBJECTID *	cqi	modulation	code_rate	sinr	efficiency	
4	4	4 QPSK	308	0	0.6016	
5	5	5 QPSK	449	1	0.877	
6	6	6 QPSK	602	2	1.17758	
7	7	7 16QAM	378	5	1.4766	
8	8	8 16QAM	490	6	1.9141	
9	9	9 16QAM	616	8	2.4063	
10	10	10 64QAM	466	10	2.7305	
11	11	11 64QAM	567	12	3.3223	
12	12	12 64QAM	666	14	3.9023	

Cqi5G						
Field: Add Calculate Selection: Select By Attributes						
OBJECTID *	cqi	modulation	code_rate	sinr	efficiency	
4	4	4 QPSK	308	-0.75	1.477	
5	5	5 QPSK	449	1.25	1.914	
6	6	6 QPSK	602	2.75	2.406	
7	7	7 16QAM	378	5	2.731	
8	8	8 16QAM	490	6.75	3.322	
9	9	9 16QAM	616	8.5	3.902	
10	10	10 64QAM	466	10.75	4.523	
11	11	11 64QAM	567	12.5	5.115	